

BUSINESS ANALYTICS PROJECT REPORT

Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)

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Week	Week 3
Subject	Business Analytics
Task Name	Multi-Echelon Supply Chain Network Design via MILP

Task Description

Students will model a global logistics network with factories, warehouses, and customer zones. They must write formal linear programming constraint matrices representing multi-echelon transport limits, warehouse storage capacities, and minimum service level bounds, solving the optimization problem to minimize total supply chain operating expenditures.

1. Project Objective

The objective is to formulate a multi-echelon supply chain network as a Mixed-Integer Linear Programming problem. The model represents flows from factories to warehouses and from warehouses to customer zones while enforcing transportation, storage, capacity, and service-level constraints.

The optimization objective is to minimize total operating expenditure while satisfying the required supply chain constraints.

2. Network Structure and Sets

The network contains three principal layers: factories, warehouses, and customer zones. Products move through the network from factories to warehouses and then from warehouses to customer zones.

Sets:

F = set of factories

W = set of warehouses

C = set of customer zones

Typical flow:

Factory f $\rightarrow$ Warehouse w $\rightarrow$ Customer zone c

The supplied task does not specify the number of factories, warehouses, customer zones, product types, costs, capacities, or service-level percentages. These are therefore modeled as configurable input parameters rather than fixed numerical values.

3. Decision Variables

Continuous shipment variables represent the quantity transported along each network link. Binary variables can be introduced when facility-opening or route-activation decisions are required.

x_{fw} = quantity shipped from factory f to warehouse w
 y_{wc} = quantity shipped from warehouse w to customer c

z_w = binary variable for warehouse activation
 $z_w \in \{0,1\}$

$x_{fw} \geq 0$
 $y_{wc} \geq 0$

The continuous variables control product flows, while binary variables allow the MILP formulation to represent discrete facility or operational decisions.

4. Objective Function

The objective is to minimize the total supply chain operating expenditure. A general formulation can include factory-to-warehouse transportation cost, warehouse-to-customer transportation cost, warehouse operating cost, and optional fixed facility costs.

Minimize:

$$\begin{aligned} Z = & \\ & \sum_{f \in F} \sum_{w \in W} c_{fw} x_{fw} \\ & + \sum_{w \in W} \sum_{c \in C} c_{wc} y_{wc} \\ & + \sum_{w \in W} h_w y_w \\ & + \sum_{w \in W} FC_w z_w \end{aligned}$$

where:

c_{fw} = factory-to-warehouse unit transport cost
 c_{wc} = warehouse-to-customer unit transport cost
 h_w = warehouse operating/handling cost
 FC_w = optional fixed warehouse cost

The exact cost components should be defined using the project dataset. The fixed-cost term is optional if warehouse activation decisions are not included.

5. Factory Capacity and Multi-Echelon Flow Constraints

Factory capacity limits ensure that total outbound shipment from each factory does not exceed its available production capacity.

Factory capacity:

$$\sum_{w \in W} x_{fw} \leq \text{Capacity}_f$$

for every factory $f \in F$

Warehouse flow balance:

$$\sum_{f \in F} x_{fw} = \sum_{c \in C} y_{wc}$$

for every warehouse $w \in W$

The flow-balance constraint ensures that, under the basic no-inventory formulation, the quantity entering a warehouse equals the quantity leaving it.

6. Warehouse Storage Capacity

Warehouse capacity constraints limit the amount of material handled or stored at each warehouse. A simple aggregate formulation is shown below.

Warehouse capacity:

$$\sum_{c \in C} y_{wc} \leq \text{StorageCap}_w \quad \forall w \in W$$

If warehouse activation is not modeled:

$$\sum_{c \in C} y_{wc} \leq \text{StorageCap}_w$$

This constraint prevents the optimization model from routing more volume through a warehouse than its available capacity allows.

7. Customer Demand and Minimum Service Levels

Customer zones require sufficient supply. A demand constraint can require the delivered quantity to meet customer demand. A minimum service-level bound can require a specified percentage of demand to be fulfilled.

Demand/service constraint:

$$\sum_{w \in W} y_{wc} \geq SL_c \times \text{Demand}_c$$

for every customer zone $c \in C$

where:

Demand_c = demand of customer zone c
 SL_c = minimum service-level fraction

For full demand fulfillment:

$SL_c = 1.00$

For example, a 95% minimum service level:

$SL_c = 0.95$

The supplied task requires minimum service-level bounds but does not provide a numerical service percentage. The percentage should therefore be supplied as an input parameter.

8. Formal Constraint Matrix Representation

The optimization problem can be represented in standard MILP form as:

```
Minimize      c■x

Subject to    A x <= b
              Aeq x = beq
              x >= 0
              z ∈ {0,1}
```

Where:

```
x  = vector of shipment and decision variables
c  = vector of operating costs
A  = inequality constraint matrix
b  = capacity/service right-hand-side vector
Aeq = equality constraint matrix
beq = flow-balance right-hand-side vector
```

The matrix form provides a compact representation suitable for MILP solvers. Rows can represent factory capacities, warehouse capacities, customer service requirements, and other operational limits.

9. Optimization Workflow and Solver

```
Input network data
|
v
Define factories, warehouses, customers
|
v
Load demand, capacity and cost parameters
|
v
Create decision variables
|
v
Build objective function
|
v
Add capacity constraints
|
v
Add flow-balance constraints
|
v
Add service-level constraints
|
v
Solve MILP
|
v
Analyze optimal flows and total cost
|
v
Generate supply chain report
```

The model can be implemented with a mathematical optimization library and a suitable MILP solver. The specific software or solver is not prescribed in the task description.

10. Results, Validation and Deliverables

Result analysis: The solution should report the selected warehouse decisions, factory-to-warehouse flows, warehouse-to-customer flows, customer service levels, capacity utilization, and minimized total operating expenditure.

Validation: Check that no factory capacity or warehouse capacity is exceeded, flow balances hold at each warehouse, minimum service-level requirements are satisfied, all shipment quantities are non-negative, and binary decisions satisfy their domain constraints.

Expected Deliverables:

1. Supply chain network formulation.
2. Decision-variable definitions.
3. MILP objective function.
4. Formal constraint matrices.
5. Factory, warehouse, and customer constraints.
6. Minimum service-level formulation.
7. Solver implementation and optimization output.
8. Cost and capacity utilization report.
9. Final analysis and conclusion.

Conclusion

This Week-3 Business Analytics task models a multi-echelon logistics network using Mixed-Integer Linear Programming. The formulation connects factories, warehouses, and customer zones through flow variables and formal constraints for transportation capacity, warehouse storage, and minimum service levels. The optimization seeks a feasible network configuration that minimizes total supply chain operating expenditures.